

Humana-Mays Healthcare Analytics2023 Case Competition

ONCOLOGY: ADE Therapy Discontinue Analytics & Predictive Modeling

(Early-stage Non-Small Cell Lung Cancer with EGFR Mutation:
Osimertinib Treatment Concerns)

October 14th, 2023

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1. Executive Summary

1.1 Study proposal

As with many other oncology drugs, tolerance of Osimertinib can be difficult due to the side effects associated with the medication—specifically nausea, fatigue, pain, high blood glucose, and constipation. Many of these side effects are manageable with proper counseling and avoidance techniques, but many patients may opt to discontinue their treatment rather than seeking guidance on how to manage them. Approximately one quarter of Humana members taking Osimertinib have side effects and discontinue their Osimertinib therapy within the first 6 months.

To address this problem, we need to leverage our data and analytics to train a ML model to accurately identify target members at risk, encourage medication adherence, and allow our oncology patients to live longer, fulfilling lives.

1.2 Modeling Summary

In our modeling process, we considered several machine learning models and found that the XG Boost Classifier outperformed the rest. After hyperparameter tuning, the model achieved a 97% AUC score on the test set, indicating its effectiveness in predicting therapy discontinuation. Extensive feature engineering on RX-Claims and Med-Claims datasets provided a comprehensive modeling dataset, and merging the datasets ensured completeness. Handling null values and aggregating features further enhanced the dataset. The model is a powerful tool for healthcare professionals and policymakers to predict and address therapy discontinuation, ultimately improving patient outcomes.

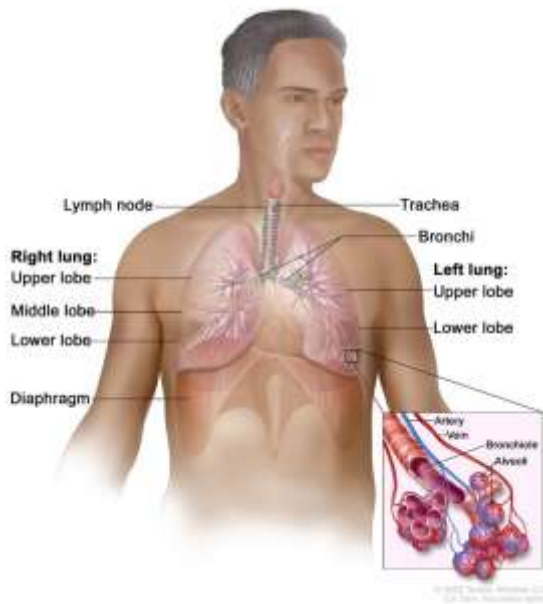
1.3 Recommendation Summary

1. **Exploratory Data Analysis:** Investigate and address gender and racial disparities in therapy discontinuation, improve treatment location coordination, understand complex interactions, mitigate financial barriers, and enhance disability and gender dynamics.
2. **Hypothesis Analysis:** Implement strategies for disease progression monitoring, alleviate financial stress, prioritize quality of life, address logistical challenges, offer social and emotional support, provide psychological counseling, improve communication, tailor treatment for comorbidities, and inform patients about clinical trials.
3. **Modeling Results:** Recognize the temporal aspect's significance in therapy discontinuation using the "tr_lag_days_tsd_rxsd_max" variable. Utilize predictive power for early intervention and develop data-driven strategies, combining patient attributes and temporal dynamics for a holistic approach.

2Case Background

2.1 Context

Oncology, a critical field of medicine, has seen remarkable advancements but remains challenged by cancer's high mortality rate, claiming nearly 600,000 lives annually in the US. Effective treatments are essential, yet ensuring patient adherence to life-saving medications is equally crucial.



Osimertinib, an oral tyrosine kinase inhibitor designed for early-stage lung cancer patients with the EGFR mutation, has proven highly effective. Patients adhering to the regimen are twice as likely to survive, with an 80% lower risk of cancer recurrence. However, Osimertinib can induce challenging side effects, including nausea, fatigue, pain, high blood glucose, and constipation.

Lung Cancer Prevalence: Lung cancer is one of the most common cancer types in the United States, and non-small cell lung cancer (NSCLC) accounts for the majority of lung cancer cases. Most lung cancer statistics include both small cell lung cancer (SCLC) and non-small cell lung cancer (NSCLC). In general,

about 10% to 15% of all lung cancers are SCLC, and about 80% to 85% are NSCLC.



Figure 1: In types of Lung Cancer NSCLC dominates

EGFR Mutation: A subset of NSCLC cases is associated with specific genetic mutations, such as those in the epidermal growth factor receptor (EGFR) gene. These mutations can drive the growth of cancer cells and are targetable with certain therapies, including drugs like Osimertinib.

EGFR-Positive Lung Cancer: In recent years, the identification of specific mutations like EGFR has led to significant advancements in the treatment of NSCLC. Targeted therapies, including EGFR tyrosine kinase inhibitors (TKIs) like Osimertinib, have improved outcomes for patients with EGFR-positive NSCLC.

Survival Rates: The use of targeted therapies like Osimertinib has led to improved survival rates for patients with EGFR-positive NSCLC. Survival rates can vary depending on factors like the stage at diagnosis and other individual characteristics.

Adherence Challenges: Despite the effectiveness of these targeted therapies, challenges related to medication adherence and the management of side effects, as mentioned in the initial context, can impact the treatment's overall success.

2.2 Problem Statement

The core problem at hand is how to address the issue of patient adherence to Osimertinib, a medication with proven life-saving potential, while ensuring effective management of its associated side effects. Non-adherence to Osimertinib treatment within the first six months, especially among those who experience side effects, presents a significant challenge. It not only puts patients at risk but also underutilizes the life-saving potential of this medication.

To resolve this problem, there is a need to leverage data and analytics to identify and target Humana members at risk of discontinuing Osimertinib treatment due to side effects. By implementing strategies to encourage medication adherence, provide proper counseling, and support patients in managing side effects, we aim to enable our oncology patients to lead longer, healthier, and more fulfilling lives. The challenge lies in developing an effective data-driven intervention plan that identifies at-risk individuals, provides them with the necessary guidance, and ultimately improves adherence rates and health outcomes for Osimertinib user

3. Preliminary Research

Lung cancer is a prevalent and significant health concern in the United States. It ranks as the second most common cancer in both men and women, excluding skin cancer. While prostate cancer is more common in men, breast cancer is more common in women. In 2023, the American Cancer Society estimates approximately 238,340 new cases of lung cancer, with 127,070 associated deaths. The disease primarily affects older individuals, with the majority of diagnoses occurring after the age of 65, and the average age at diagnosis is around 70.

Regrettably, lung cancer is the leading cause of cancer-related deaths in the U.S., accounting for about one-fifth of all cancer fatalities. Annually, more lives are claimed by lung cancer than by colon, breast, and prostate cancers combined.

On a positive note, there is a declining trend in new lung cancer cases, largely attributed to more people quitting smoking or never starting. Advances in early detection and treatment have also contributed to the reduction in lung cancer-related deaths.

The lifetime risk of developing lung cancer varies, with an approximate 1 in 16 chance for men and 1 in 17 for women. These risks encompass both smokers and non-smokers, with significantly higher risks for those who smoke. Interestingly, disparities exist between Black and White populations, with Black men having a 12% higher likelihood of developing lung cancer than White men, while the rate is 16% lower in Black women compared to White women. The gender gap in lung cancer rates is diminishing, with rates dropping in men over recent decades and, more recently, in women.

Lung cancer's prognosis is generally serious, but some individuals with earlier-stage cancers achieve cure. Survival rates vary depending on factors like the type of lung cancer, stage at diagnosis, and other variables.

4. Analysis, Insights and Recommendations

4.1 Data Set and Goal of the Analysis

Data Sets:

Train Data:

- **Target.csv:** This dataset contains 1,232 observations and encompasses 10 variables.
- **RX_Claims.csv:** With 32,133 observations and 24 variables, this dataset provides extensive information on medical claims related to prescription medications.
- **Med_Claims.csv:** With a whopping 100,159 observations and 27 variables, this dataset offers a comprehensive view of medical claims data.

Hold-Out Data:

- **Target.csv:** Comprising 420 observations and 8 variables, this dataset serves as a valuable comparison to the train data.
- **RX_Claims.csv:** With 6,669 observations and 24 variables, this dataset is essential for evaluating the model's performance on new data.
- **Med_Claims.csv:** This dataset contains 23,232 observations and 27 variables, mirroring the complexity of the train data.

NOTE:

All the analyses and insights provided in this analysis were derived exclusively from the Train datasets. The Train datasets served as the foundation for developing an understanding of the target variable and refining our approach to the analysis.

Goal or Target Analysis: "tgt_ade_dc_ind"

Objective: The primary objective is to uncover patterns and insights related to patients discontinuing ADE (Adverse Drug Event) therapy.

Key Questions: To achieve this objective, we must address several key questions:

- **Why are patients discontinuing?** Investigate the underlying reasons driving therapy discontinuation.
- **What are the common reasons?** Identify the most frequently occurring factors contributing to discontinuation.
- **How can we reduce discontinuation rates?** Explore strategies and interventions to mitigate discontinuation.

- **What recommendations and action plans can be implemented?** Develop actionable solutions based on the findings.
- **What predictive modeling techniques can be used?** Employ predictive modeling to gain insights and make informed decisions.

Conclusion: In conclusion, the analysis will culminate in a comprehensive summary of the findings. This summary will not only encapsulate the insights gained but also present actionable recommendations based on the analysis. These recommendations are vital for addressing discontinuation rates and improving patient outcomes during ADE therapy.

4.2 Diagnostic, Descriptive and Exploratory Analysis:

Target Variable - "tgt_ade_dc_ind":

The target variable, "tgt_ade_dc_ind," serves as a critical indicator within our dataset. It provides insights into whether an individual reported an Adverse Drug Event (ADE) and subsequently discontinued their therapy before its completion. This variable holds essential information for our analysis.

ADE Observations:

Understanding the ADE occurrences in our dataset is pivotal:

- **Patients with ADE (1):** Out of a total of 1,232 patients, 117 experienced an Adverse Drug Event (ADE). These cases are marked with a "1" under the "tgt_ade_dc_ind" variable, signifying the occurrence of an ADE during their therapy.
- **Patients without ADE (0):** The majority of patients, precisely 1,115 out of 1,232, did not encounter an ADE throughout their 6-month Osimertinib therapy. These cases are identified with a "0" in the "tgt_ade_dc_ind" variable, indicating the absence of an ADE event during their therapy.

Gender Breakdown of Patients Who Successfully Completed ADE Therapy:

Understanding the gender distribution of patients who completed ADE therapy successfully is valuable:

- **Female Patients without ADE:** Among patients who completed their ADE therapy successfully, 730 were females. These individuals did not experience an ADE during their therapy.
- **Male Patients without ADE:** Similarly, 303 male patients successfully finished their ADE therapy without encountering any ADE events.
- **Patients without ADE with Unknown Gender:** Additionally, 82 patients who completed their ADE therapy without any ADE occurrences had unspecified or unknown gender information.

Gender Breakdown of Patients Who Discontinued ADE Therapy:

It's essential to examine the gender distribution of patients who discontinued their ADE therapy:

- **Female Patients with ADE in the First 6 Months:** Among patients who discontinued their ADE therapy, 85 were female. These individuals experienced an ADE event within the first 6 months of their therapy.
- **Male Patients with ADE in the First 6 Months:** A total of 31 male patients discontinued their ADE therapy after encountering an ADE within the first 6 months.

Understanding the interplay between ADE occurrences, therapy completion, and gender provides a comprehensive view of patient outcomes. This detailed information aids in drawing meaningful conclusions and tailoring interventions to improve therapy adherence and overall patient well-being.

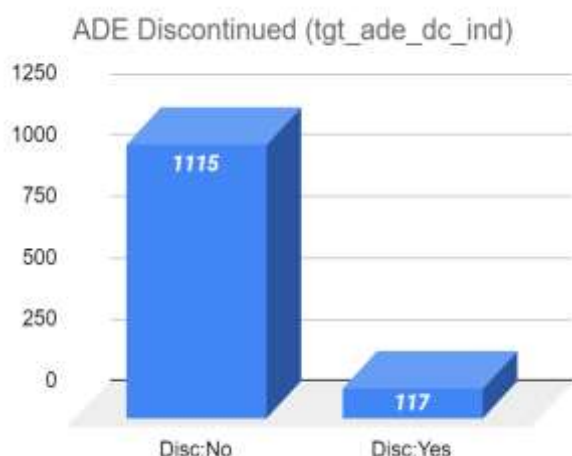


Figure 2: Overall therapy Completed & Discontinued

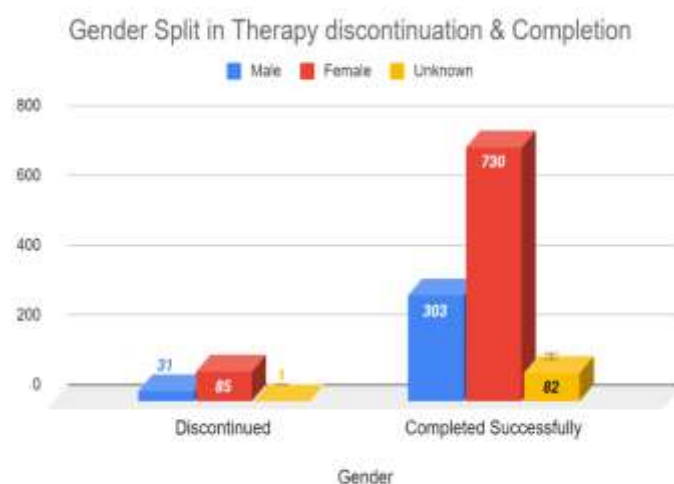


Figure 3: Gender Split in Therapy discontinuation & Completion

Insights on Discontinuation Rate by Gender:

Analyzing the discontinuation rate based on gender provides valuable insights into therapy outcomes:

- **Male Discontinuation Rate:** The discontinuation rate among male patients is 10%. This percentage represents the proportion of male patients who discontinue therapy, as indicated by the available data.
- **Female Discontinuation Rate:** In contrast, the discontinuation rate among female patients is slightly higher at 11.6%. This figure reflects the percentage of female patients who discontinue therapy.

Gender Disparity in Discontinuation Rates:

It is noteworthy that, according to the data provided, there is a gender disparity in therapy discontinuation rates:

- **Female Discontinuation:** Females exhibit a higher discontinuation rate of 11.6%. This suggests that, on average, a larger proportion of female patients discontinue therapy compared to their male counterparts.

Exploring the reasons behind this gender disparity in discontinuation rates can provide critical insights. Understanding these differences can lead to tailored interventions to address the specific needs of different patient groups and enhance overall therapy adherence and effectiveness.

Race Breakdown on Osimertinib ADE therapy

Race Categories/Codes:

To interpret the race-related data effectively, we use the following race categories and corresponding codes:

- 0 = Unknown
- 1 = White
- 2 = Black
- 3 = Other (unspecified races)
- 4 = Asian
- 5 = Hispanic
- 6 = Native American (North American native)

Race Breakdown of All Patients:

Understanding the distribution of races among all patients provides essential context:

- **Unknown Race:** There are 152 patients for whom the race is categorized as unknown.
- **White Patients:** The dataset includes 699 White patients.
- **Black Patients:** There are 145 Black patients.
- **Other Unspecified Races:** The dataset contains 37 patients classified under "Other," representing unspecified racial backgrounds.
- **Asian Patients:** The dataset includes 152 Asian patients.
- **Hispanic Patients:** There are 42 Hispanic patients.
- **Native American Patients:** A total of 5 patients are classified as Native American.
- **Total Patients:** The dataset encompasses a total of 1,232 individuals.

Race Breakdown of Patients Who Discontinued Osimertinib ADE Therapy:

Analyzing the racial breakdown of patients who discontinued the therapy yields important insights:

- **Unknown Race with ADE:** Among the patients who discontinued Osimertinib therapy and had an adverse drug event (ADE), 7 fall under the unknown race category.

- **White Patients:** 74 White patients discontinued Osimertinib therapy.
- **Black Patients:** 14 Black patients discontinued Osimertinib therapy.
- **Other Unspecified Races:** 5 patients classified under "Other" discontinued Osimertinib therapy.
- **Asian Patients:** 9 Asian patients discontinued Osimertinib therapy.
- **Hispanic Patients:** 7 Hispanic patients discontinued Osimertinib therapy.
- **Native American Patients:** 1 Native American patient discontinued Osimertinib therapy.

Understanding the racial distribution among patients who discontinued the therapy is crucial for tailoring interventions and addressing any disparities in therapy discontinuation rates across different racial groups. See the interesting finding in insights section.

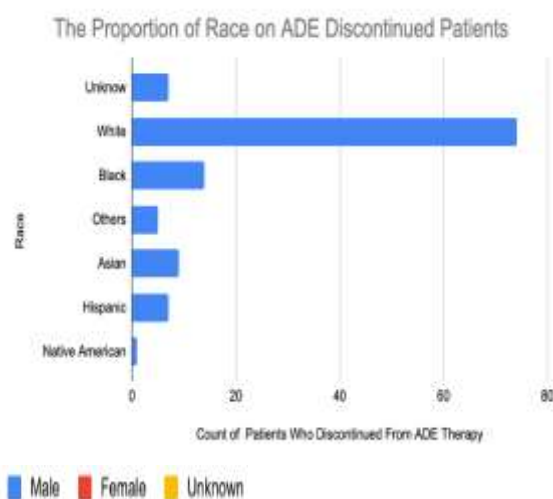


Figure 4: Racial distribution of ADE discontinued patients

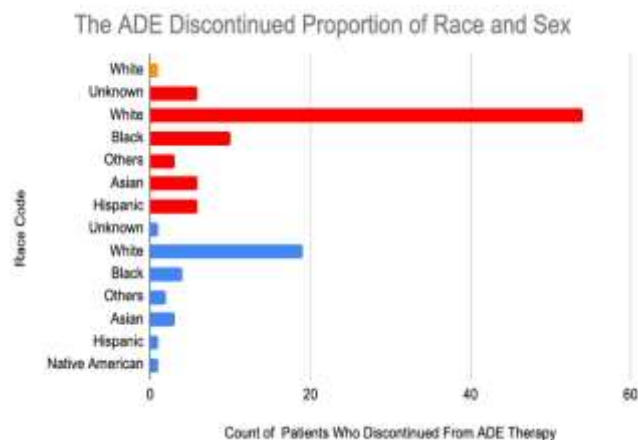


Figure 5: Race Vs Sex Vs ADE Discontinuation

Insights on Race:

Discontinuation Rate by Individual Race Contribution:

Understanding the discontinuation rate across various racial categories is crucial. Let's delve into the specifics:

- **Unknown:** The discontinuation rate for individuals with an unknown race is 4%. This category comprises cases where the patient's race is not specified, making it essential to examine further.
- **White:** Among White patients, the discontinuation rate is 10%. This racial group constitutes a significant portion of the dataset, and a 10% discontinuation rate reflects the specific challenges faced by White patients.

- **Black:** The discontinuation rate for Black patients is 9.6%, indicating a slightly lower rate compared to White patients. It's valuable to explore the factors contributing to this variation.
- **Other:** The discontinuation rate among individuals categorized as 'Other' is 13%. This group comprises races not mentioned separately, and the higher discontinuation rate warrants a closer look.
- **Asian:** The discontinuation rate for Asian patients stands at 5%, indicating a relatively lower rate compared to some other racial categories.
- **Hispanic:** Hispanic patients exhibit a discontinuation rate of 16%, suggesting a higher rate compared to most other racial groups. Understanding the factors behind this higher discontinuation rate is essential.
- **Native American:** Native American patients have the highest discontinuation rate at 20%, indicating a significant challenge in this racial category. Further analysis is crucial to address this issue effectively.

In conclusion, analyzing the discontinuation rate across various racial categories is essential to identify disparities and develop strategies to improve the discontinuation rate, especially among racial groups with higher rates.

Estimated Age Breakdown on Osimertinib ADE therapy

Age in the Target Train Dataset:

- The age of patients in the Target Train Dataset spans from **38 to 96** years.
- The youngest patient in the dataset is 38 years old, while the oldest is 96 years old.

Frequency of Estimate Age:

- Notably, the most common age among patients in this dataset is **65 years**, with a total of 68 individuals at this age.
- Conversely, the dataset contains less frequent ages, such as 48, 39, 38, 96, and 95, each occurring only once.

Age and Lung Cancer:

- An interesting pattern emerges when examining the relationship between age and lung cancer. The data indicates that lung cancer is most prevalent among patients aged **65 to 86**, after which its prevalence gradually decreases.

Unknown Estimate Age:

- It's essential to note that there are 83 entries in the dataset with an unknown or null estimate age value. These entries should be treated as a distinct category in our analysis.

Average and Median Age:

- The average age of patients who discontinued ADE therapy is approximately 73 years.
- Similarly, the median age of patients who discontinued ADE therapy is also approximately 73 years.

HEDIS(POT) Breakdown on Discontinued from Osimertinib ADE therapy

Impact of Place of Treatment on ADE Therapy Discontinuation:

The choice of a different Place of Treatment (HEDIS) by patients during Adverse Drug Event (ADE) therapy can have a substantial impact, particularly for patients who selected four different Places of Treatment. This choice has been linked to a higher likelihood of discontinuing ADE therapy.

Key Consideration:

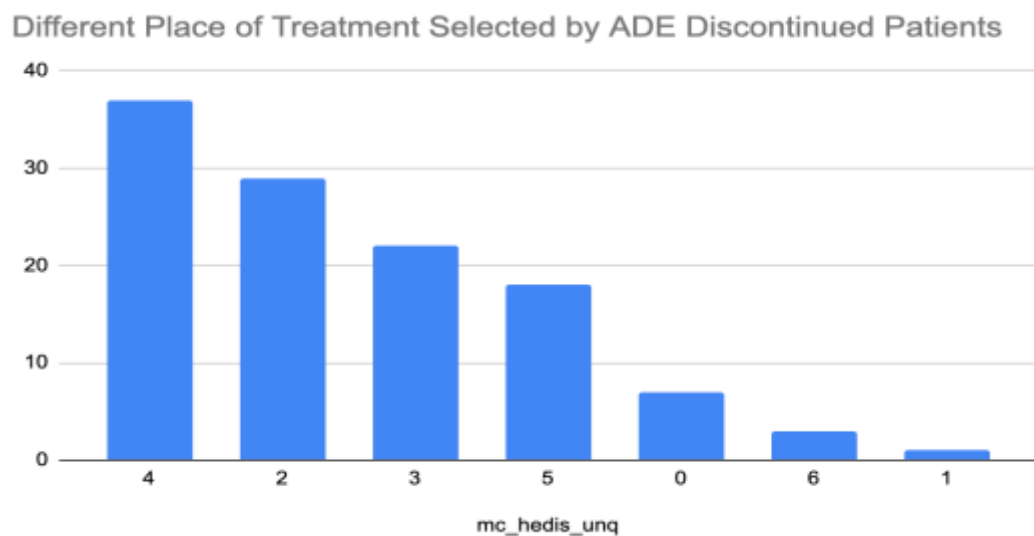
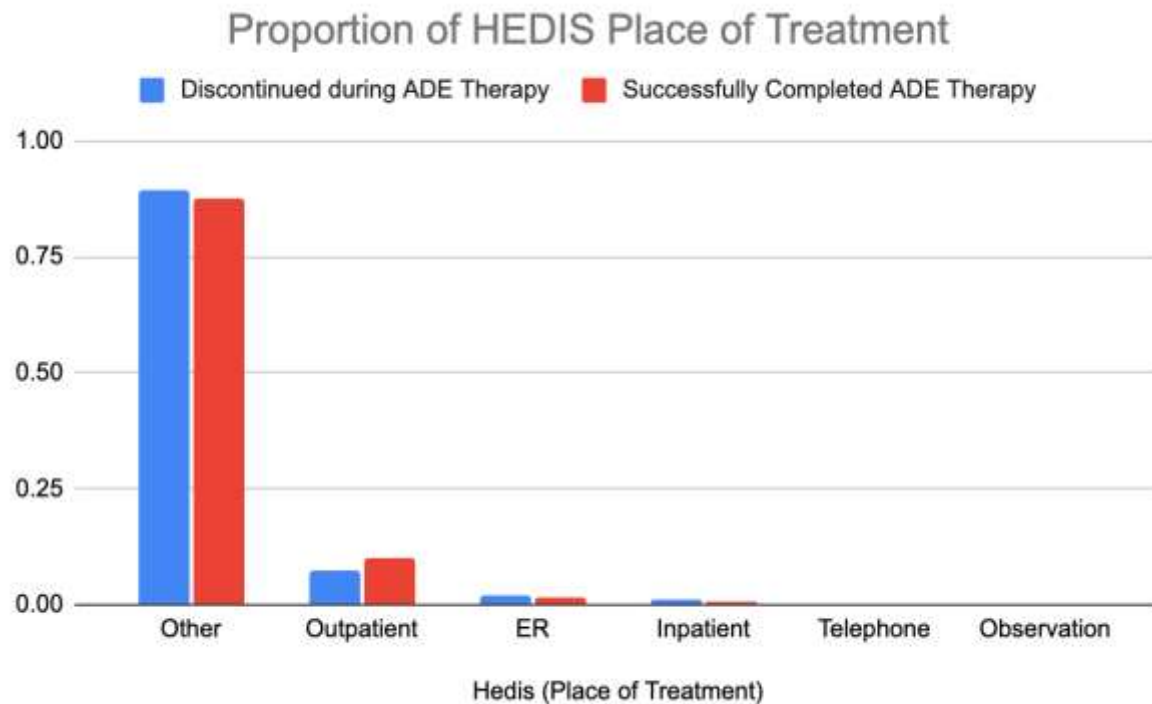
The selection of different Places of Treatment during ADE therapy is a critical factor that significantly influences the therapy's continuity. It is crucial to explore this aspect further to understand the reasons behind this trend and its implications on patient outcomes.

Elaboration:

Patients who opt for multiple and varying Places of Treatment may face challenges in terms of consistency and coordination of care. This can result in disruptions to their treatment regimen and may lead to a higher likelihood of discontinuation. Factors such as travel distance, communication between healthcare providers, and the availability of necessary resources at different treatment locations can all play a role in this dynamic.

To gain a more comprehensive understanding of this issue, it's essential to investigate the specific reasons behind patients' choices, the extent of discontinuation, and whether there are opportunities to improve the coordination and experience for patients who select multiple Places of Treatment during their ADE therapy.

By delving deeper into this analysis and addressing the underlying factors, healthcare providers and policymakers can develop strategies to enhance patient experiences and reduce the likelihood of therapy discontinuation, ultimately improving patient outcomes during ADE therapy.



Insights on HEDIS POT:

Two charts have been provided, each shedding light on different aspects related to patient therapy discontinuation. Let's delve into the insights derived from these charts.

Chart 1: Proportion of HEDIS Place of Treatment:

- This first chart, illustrating the "proportion of the HEDIS place of treatment," doesn't exhibit a direct impact on patient discontinuation rates. The data presented here provides valuable information about the distribution of treatment locations but doesn't offer immediate insights into why patients decide to discontinue therapy.

Chart 2: Different Place of Treatment Selected by ADE Discontinued Patients:

- In contrast, the second chart, which highlights "different places of treatment selected by ADE-discontinued patients," presents a more compelling narrative. It suggests that the choice of treatment location during therapy has a role in influencing patients' decisions to discontinue ADE therapy.
- Patients actively choosing various types of treatment locations during their therapy journey is a crucial observation. By conducting a chain relation analysis based on this data, it's possible to develop a super linear score-based metric. This metric can help predict when patients are likely to discontinue therapy while they are still in the treatment process.
- Implementing this predictive modeling approach at the production level, particularly at the stream level, can be highly beneficial. It would enable healthcare professionals to proactively identify patients at risk of discontinuing therapy, allowing for timely interventions and a more tailored approach to patient care.

In summary, the second chart holds significant potential in enhancing our understanding of therapy discontinuation factors and provides a foundation for predictive modeling to improve patient adherence and overall therapy outcomes. Further analysis and implementation of these insights have the potential to transform patient care in a proactive and data-driven manner.

Side-Effects which Reported on Claims Breakdown on Discontinued from Osimertinib ADE therapy:

Impact of Side Effects on ADE Therapy Discontinuation:

The presence of side effects reported on claims during Adverse Drug Event (ADE) therapy has a significant impact on the discontinuation rate of the ADE therapy. It's essential to understand the relationship between side effects and patient decisions to discontinue therapy.

Key Side Effects:

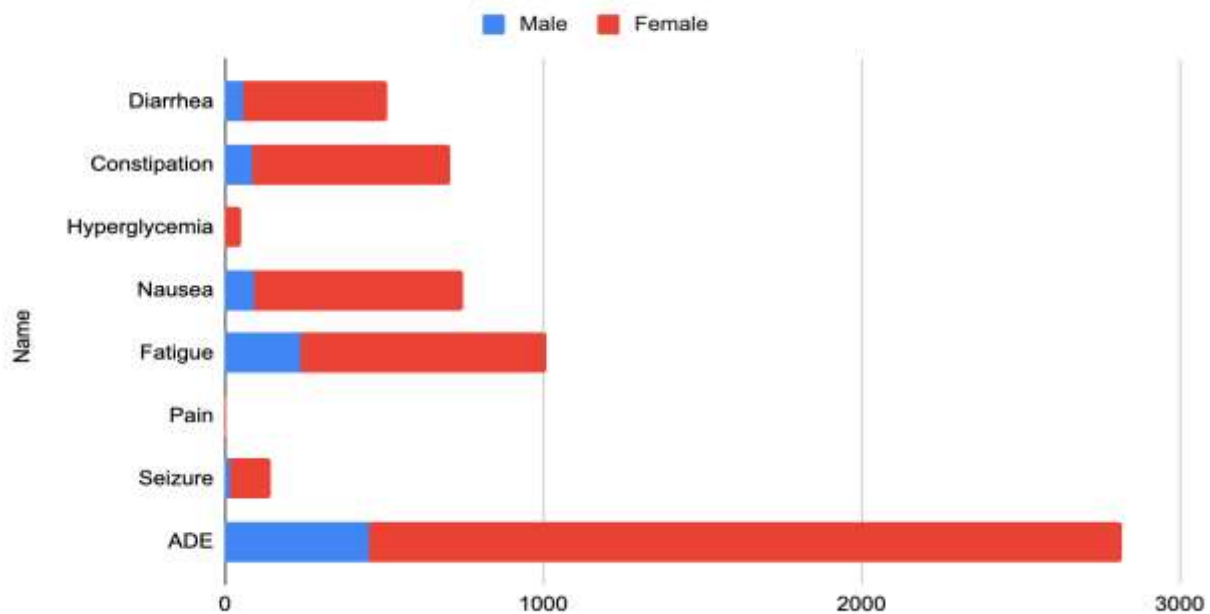
Among the various reported side effects, four of them stand out as the top contributors to therapy discontinuation during Osimertinib ADE therapy: Fatigue, Nausea, Constipation, and Diarrhea. These side effects have been observed to have a pronounced influence on patients' decisions to discontinue their therapy.

Additional Information Required:

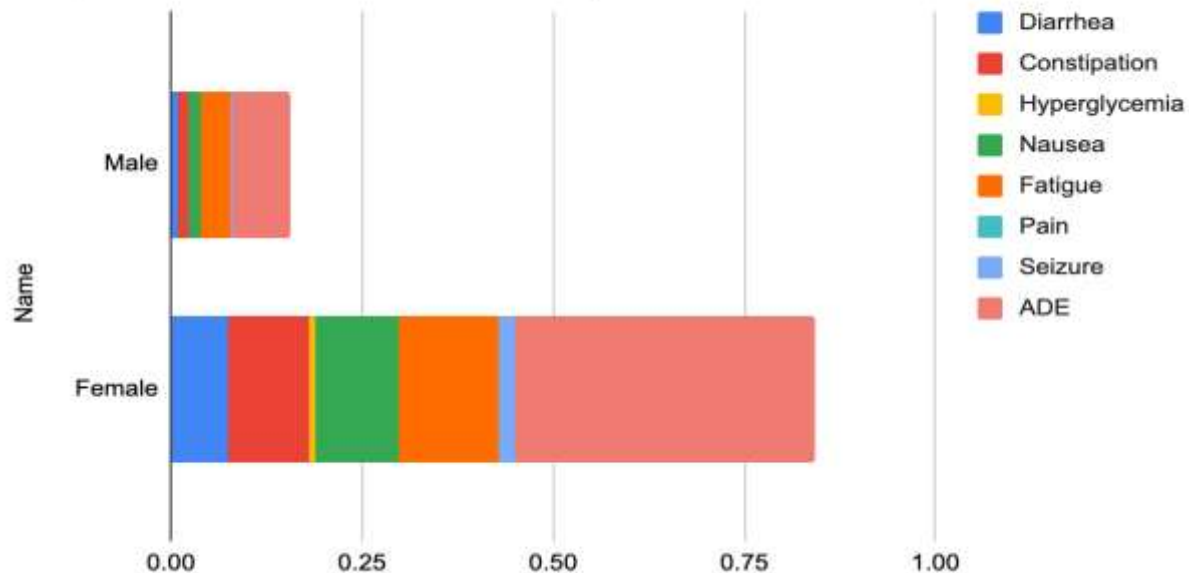
To delve deeper into this analysis and gain a more comprehensive understanding, we need to consider several critical factors:

- **Severity Rate:** It's crucial to assess the severity rate of these four prominent side effects. Understanding the degree of severity associated with each side effect can provide insights into the impact they have on patients' well-being and their decisions to discontinue therapy.
- **Mortality Rate:** An essential consideration is whether patients who experienced these specific side effects during therapy ultimately passed away. Assessing the correlation between these side effects and patient mortality can shed light on the gravity of the situation.

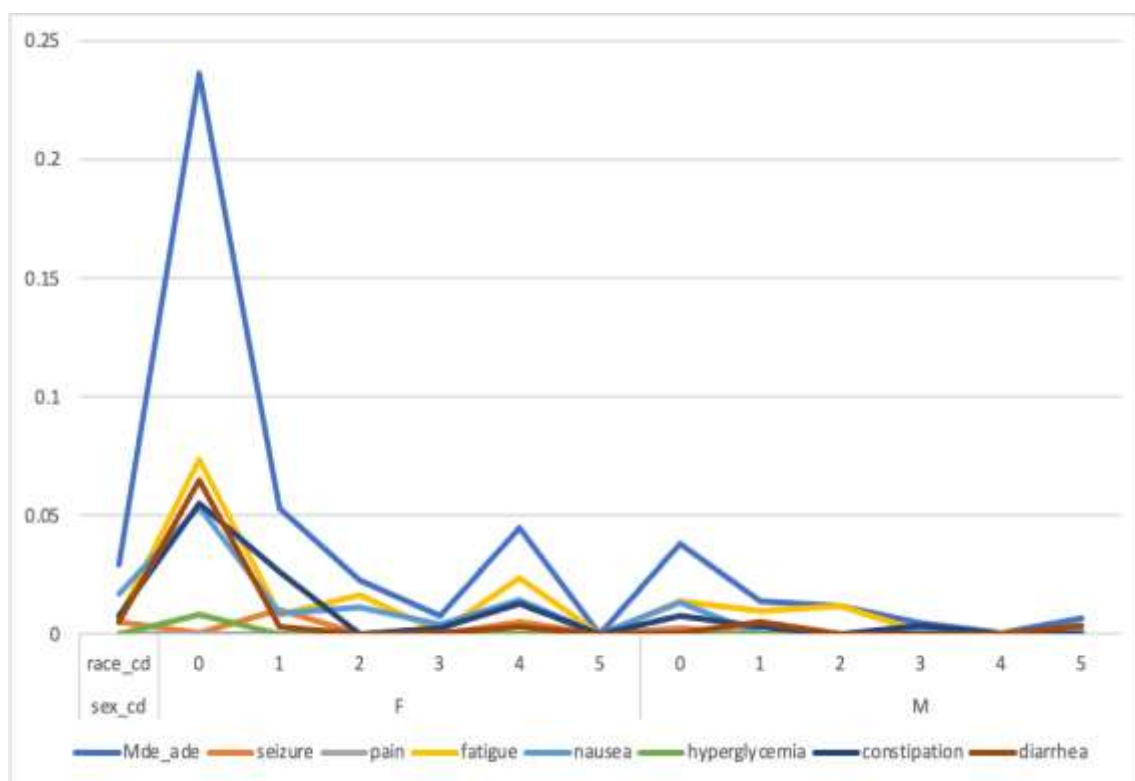
SideEffects which Reported on Claims - MedClaims



Proportion of SideEffects which Reported on Claims -MedClaims



Proportion of Side Effects compare to Sex and Race Code



Insights on Side Effects:

Key Side Effects Leading to Discontinuation:

- Among the myriad of side effects reported, four have emerged as significant contributors to therapy discontinuation during Osimertinib ADE therapy: Fatigue, Nausea, Constipation, and Diarrhea. These specific side effects have been observed to exert a considerable influence on patients' decisions to discontinue their therapy.

Comparison of Sex and Race Code Impacts on Side Effects and Discontinuation:

In analyzing the impact of sex and race code on side effects and patient decisions to discontinue from ADE therapy, several insights have surfaced:

- **High Discontinuation Rate Among White Females:** It has been observed that among the demographic groups, White females have a notably high discontinuation rate. However, it's crucial to note that the given sample data proportionally represents White and female patients.

This observation raises questions about the interplay of factors such as race, gender, and side effects in the decision-making process regarding therapy discontinuation. It suggests that more comprehensive analysis, accounting for sample representation and the complex relationships between these variables, may be needed to understand and address the disparities in discontinuation rates among different patient groups effectively.

In conclusion, these insights illuminate the pivotal role of specific side effects in therapy discontinuation and hint at the complex interactions among sex, race, and side effects in influencing patient decisions. Further in-depth analysis is essential to develop targeted strategies and interventions for improving therapy adherence and patient outcomes.

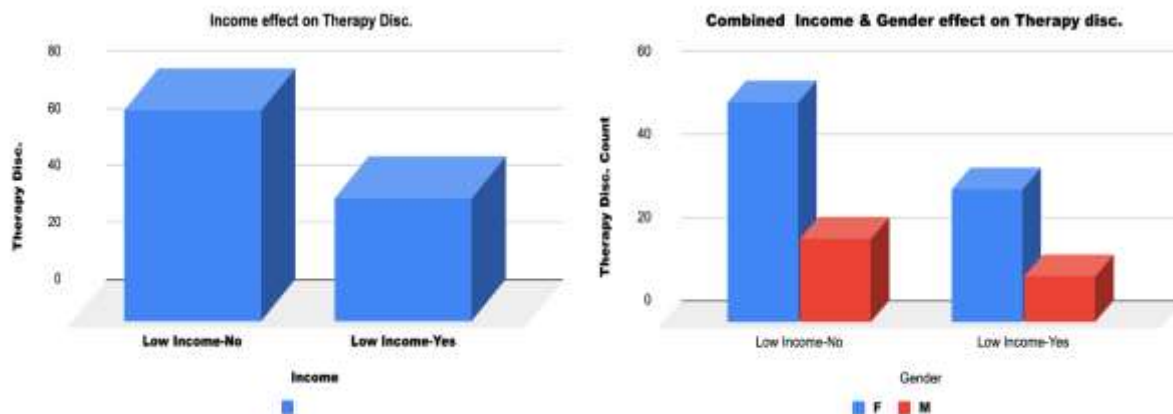
Low Income Impact on Discontinued from Osimertinib ADE therapy:

As depicted in the charts below, income indicators emerge as a pivotal factor influencing patients' decisions to discontinue ADE therapy. Notably, approximately **30%**(40/117) of patients who chose to discontinue their therapy fall into the category of low-income individuals or families.

The Role of Money/Income:

- Financial considerations undoubtedly play a crucial role here. Patients who are unable to afford the costs associated with therapy may find themselves compelled to discontinue their treatment, irrespective of their gender.
- However, a particularly intriguing and high-impact insight surfaces when examining the second chart. It delves into the combined effect of income and gender on therapy discontinuation.

The interplay between income and gender may have a significant influence on patients' decisions regarding therapy continuation. It invites further exploration into the complex relationships between these variables and the development of targeted strategies to address the disparities in therapy adherence and outcomes among different patient groups.



Insights on Low Income:

Overall Impact of Low Income:

- It's striking to note that around 30%, approximately 40 out of 117 patients, opted to discontinue their therapy, and they all share a common denominator—being classified as low-income individuals or families. This holds true across genders and serves as a central point of consideration when evaluating the relationship between income and therapy discontinuation (refer to the "Income Effect on Therapy Disc" chart for details).

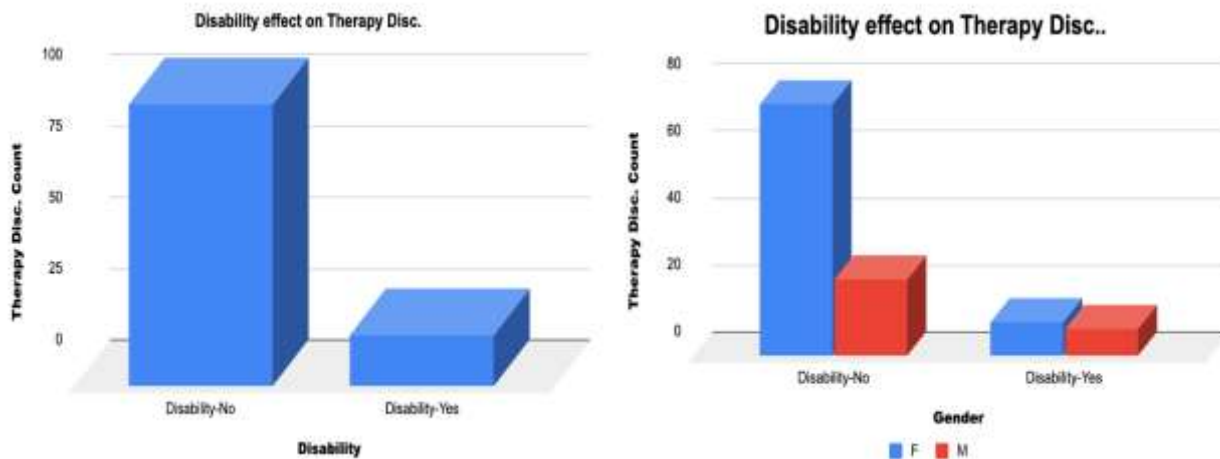
Comparing Low Income Based on Gender:

- To delve further into the impact of low income with respect to gender, it's revealing to observe that out of the total 85 female patients who discontinued from ADE therapy, approximately 37% (30 out of 85) were classified as low-income individuals or families. A similar trend is observed among male patients, with approximately 34% of the 35 male patients who discontinued falling into the low-income category.

These findings underscore the pronounced influence of income status on therapy discontinuation, cutting across gender lines. Recognizing this influence allows healthcare professionals and policymakers to tailor interventions and support systems that address the financial barriers patients face, thereby enhancing therapy adherence and ultimately improving patient outcomes.

Disability Effect on Discontinued from Osimertinib ADE therapy:

In our pursuit to understand the role of disability in patients who discontinued Osimertinib ADE therapy, we conducted a comprehensive analysis. The results offer valuable insights:



- A mere 10% of patients reported or were classified as having a disability before commencing their therapy. Based on this analysis, it appears that disability, as a standalone factor, does not significantly contribute to patients' decisions to discontinue their ADE therapy.

However, when we consider the aspect of gender, an intriguing pattern emerges:

- It's worthy that a higher percentage of male patients acknowledged having a disability indicator compared to their female counterparts. This observation implies that gender might play a more influential role in patients' responses to the disability indicator than the presence of disability itself.

Understanding these nuances in the context of disability and gender is crucial for healthcare professionals and researchers. It can lead to the development of more tailored interventions and support systems, allowing for improved therapy adherence and patient outcomes.

4.3 Analysis and Insights Summary:

Train Data Summary:

- **Target.csv:** Contains 1,232 observations and 10 variables.
- **RX_Claims.csv:** Includes 32,133 observations and 24 variables, focusing on medical claims related to prescription medications.
- **Med_Claims.csv:** Offers a comprehensive view of medical claims data with 100,159 observations and 27 variables.

- **HoldOut Data:** The holdout data is used for model evaluation and consists of three datasets with similar structures to the train data.

Goal and Objective:

- The primary objective is to uncover patterns and insights related to patients discontinuing ADE (Adverse Drug Event) therapy.
- Key questions include understanding the reasons behind discontinuation, common factors contributing to discontinuation, strategies for reducing discontinuation rates, recommendations for action plans, and the application of predictive modeling techniques.

Target Variable - "tgt_ade_dc_ind":

- This variable indicates whether an individual reported an ADE and discontinued therapy before its completion, providing crucial insights into the dataset.

ADE Observations:

- Out of 1,232 patients, 117 experienced an ADE (1), while 1,115 did not (0).

Gender and Discontinuation Rate:

- Gender breakdown shows the number of patients who completed ADE therapy and the number who discontinued based on ADE presence or absence.
- The discontinuation rate among females is 11.6%, slightly higher than the 10% rate among males.

Race Impact on Discontinuation of ADE Therapy:

- Data includes categories and codes for different races.
- It reveals the racial distribution among all patients and those who discontinued ADE therapy
- Unknown race has a 4% discontinuation rate.
- White patients have a 10% discontinuation rate.
- Black patients exhibit a 9.6% rate.
- Other unspecified races have a 13% rate.
- Asian patients have a 5% rate.
- Hispanic patients have a 16% rate.
- Native American patients have the highest rate at 20%.
- But on overall sample patient head count, race code "White" patients are contributing to 50%-60%.

Estimated Age and ADE Therapy:

- The dataset ranges from ages 38 to 96.
- The most common age is 65 years, with lung cancer peaking between ages 65 and 86.

HEDIS Place of Treatment and ADE Therapy:

- Patients choosing multiple places of treatment may have a higher likelihood of discontinuing ADE therapy.

Side Effects and Discontinuation:

- Fatigue, Nausea, Constipation, and Diarrhea are the top contributors to therapy discontinuation.
- The severity and mortality rates related to these side effects require further analysis.

Income and Discontinuation:

- Around 30% of patients who discontinued therapy were low-income individuals or families, regardless of gender.
- The interplay between income and gender may be influencing discontinuation.

Disability and Discontinuation:

- Only 10% of patients reported a disability before therapy.
- A higher percentage of male patients acknowledged a disability indicator compared to females.

Insights Recommendations:

- Understanding gender and racial disparities in discontinuation rates is essential.
- Financial barriers are significant in therapy discontinuation.
- The choice of multiple treatment locations may impact discontinuation rates.
- Side effects play a crucial role in therapy discontinuation.
- Further exploration is needed to address these complex relationships and improve therapy adherence and patient outcomes.

4.4 Recommendation on Analysis Performed:

1. **Address Gender Disparities in Therapy Discontinuation:**
 - Explore the reasons behind gender-based disparities in therapy discontinuation rates.
 - Tailor interventions to address the specific needs of different gender groups and enhance therapy adherence.
2. **Focus on Racial Disparities in Discontinuation Rates:**
 - Understand the racial distribution among patients who discontinue therapy.
 - Develop strategies to reduce discontinuation rates, particularly among racial groups with higher rates.
3. **Enhance Coordination of Treatment Locations:**
 - Investigate the reasons behind patients' choices of multiple treatment locations.

- Improve the coordination and patient experience for those selecting various Places of Treatment during ADE therapy.
- 4. Investigate Complex Interactions:
 - Examine the interplay of factors such as race, gender, and side effects in therapy discontinuation decisions.
 - Consider a comprehensive analysis to address disparities in discontinuation rates among different patient groups.
- 5. Address Financial Barriers:
 - Recognize the impact of income on therapy discontinuation.
 - Tailor interventions and support systems to mitigate financial barriers and improve therapy adherence.
- 6. Focus on Disability and Gender Dynamics:
 - Investigate the relationships between disability, gender, and therapy outcomes.
 - Develop tailored interventions and support systems to enhance therapy adherence and patient outcomes.
- 7. Implement Predictive Modeling:
 - Utilize data-driven strategies to transform patient care and proactively identify therapy discontinuation risks.

5. Modeling

5.1 Data Preprocessing:

Train Data:

- **Target.csv:** This dataset contains 1,232 observations and encompasses 10 variables.
- **RX_Claims.csv:** With 32,133 observations and 24 variables, this dataset provides extensive information on medical claims related to prescription medications.
- **Med_Claims.csv:** With a whopping 100,159 observations and 27 variables, this dataset offers a comprehensive view of medical claims data.

HoldOut Data:

- **Target.csv:** Comprising 420 observations and 8 variables, this dataset serves as a valuable comparison to the train data.
- **RX_Claims.csv:** With 6,669 observations and 24 variables, this dataset is essential for evaluating the model's performance on new data.
- **Med_Claims.csv:** This dataset contains 23,232 observations and 27 variables, mirroring the complexity of the train data.
-

Null Value Imputation for Train Dataset:

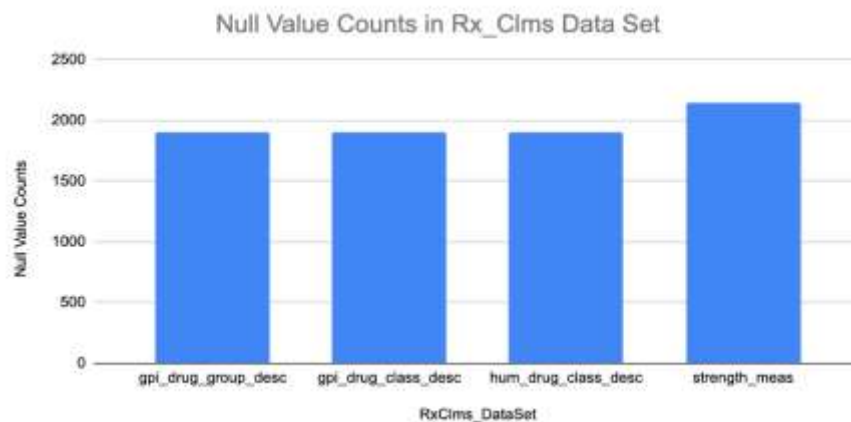
Null Value Counts in Target.csv:



-
- The variables 'Est_age,' 'cms_disabled_ind,' and 'cms_low_income_ind' have been imputed with a value of "0." This imputation is appropriate because these variables serve as indicator variables, typically taking binary values to indicate the presence or absence of certain conditions or characteristics.
- For 'Est_age,' the value "0" is used to signify that there is no specific age estimate available, which aligns with the binary nature of this variable.

- Similarly, 'cms_disabled_ind' and 'cms_low_income_ind' being indicator variables imply that "0" indicates the absence of disability or low-income status, respectively.
- On the other hand, the 'Race_cd' and 'Sex_cd' variables have been imputed with "99." This imputation strategy is suitable because these variables are categorical, meaning they represent distinct categories or classes.
- By assigning a value of "99" to these variables, it effectively creates a separate class within each variable. In the modeling process, this separate class can be treated as a distinct category, allowing the model to consider cases where the race or gender information is missing or unspecified.
- This approach ensures that no information is lost due to missing data in these categorical variables and that they can be effectively incorporated into the modeling process. It's important to note that this imputation strategy should be well-documented and communicated clearly to ensure that it's properly handled during subsequent data analysis and modeling steps.

- **Null Value Counts in RX_Claims.csv:**



- The columns mentioned are categorical in nature, and they have been imputed with the value "9999" to address missing or null values. This imputation strategy is often used in data preprocessing to ensure that the categorical variables are appropriately handled during subsequent data analysis or modeling processes.

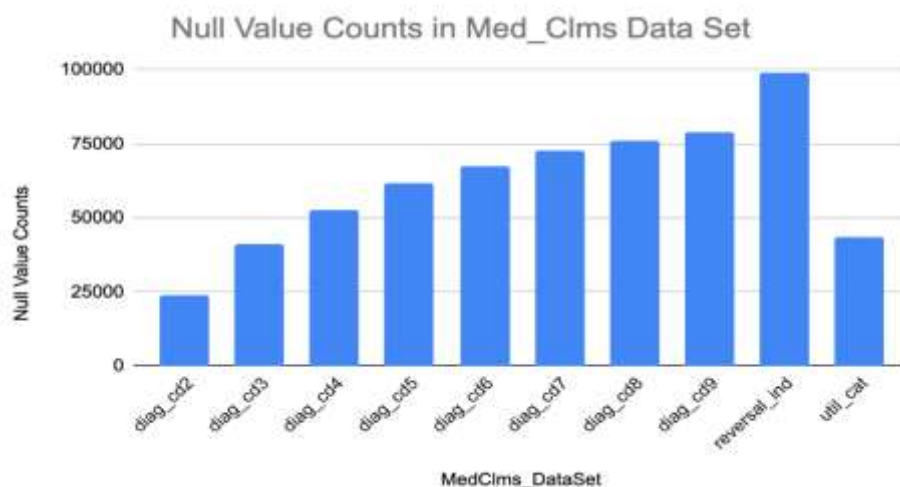
Imputing missing values with "9999" serves several purposes:

- **Preservation of Information:** By using "9999" as the imputed value, it creates a separate category or class within these categorical columns. This allows the model to consider cases where the data is missing or unspecified, ensuring that no information is lost.

- **Distinct Treatment:** When the model encounters the value "9999," it can treat it as a distinct category, ensuring that any patterns or relationships associated with missing data can be taken into account during analysis.
- **Ease of Identification:** Imputing with "9999" makes it easy to identify and handle missing values during data analysis. It's a clear and explicit way to mark data points where information is missing.

However, it's important to note that the choice of imputed value, in this case, "9999," should be well-documented and communicated clearly within the data analysis and modeling process. Additionally, any potential impact or influence of this imputation strategy on the results of the analysis should be carefully considered and evaluated. The ultimate goal is to ensure that the imputation method aligns with the objectives of the analysis and provides meaningful insights while appropriately handling missing data in the categorical columns.

- **Null Value Counts in Med_Claims.csv:**



- Categorical variables, representing distinct categories, had missing values filled with "9999."
- This approach retains all records, uses a distinct marker for missing data, and aligns with modeling techniques.
- The 'Med_Claims' dataset's numerical variables contain no null values, simplifying data handling for this specific dataset.

Value Imputation for HoldOut Dataset:

Imputation in the HoldOut Data set mirrors the process in the TrainSet, ensuring that the data is consistent and ready for analysis. The imputation techniques employed maintain the integrity of the dataset by addressing missing values effectively.

Just as in the TrainSet:

- Numerical variables in the HoldOut Data set are processed to impute null values appropriately.
- Categorical variables receive special treatment by imputing them with "9999."

This consistent approach across both datasets is vital for generating reliable and comparable results. It ensures that the HoldOut Data set is prepared for analysis in a manner that aligns with the methods applied to the TrainSet, promoting the quality and reliability of any insights or models derived from the data.

5.2 Data Manipulation and Feature Engineering:

To construct a robust model, we are working with three distinct data sets: Target.csv, Rx_Clms, and Med_Clms. Each data set contains varying observations and columns, and we aim to merge them cohesively for model development. This necessitates the identification of a common primary key that exists in all data sets or can be engineered as a new primary key to facilitate the merging process.

In our case, we've chosen "therapy_id" as the primary key for merging all three data sets. However, it's important to note that simply merging the data sets as they are would not be sufficient to build an effective model. This is where feature engineering comes into play. We need to transform and enrich each table individually before merging them into a comprehensive modeling data set.

A practical example of how we approach feature engineering is as follows:

1. Rx_Claims Feature Engineering:

- In the Rx_Claims data set, we initially have 32,000 observations with 1,160 unique therapy_ids. Our objective is to transform this data into a format that can be effectively merged with the Target data set (which has a unique therapy_id for each observation).
- We apply grouping based on the "therapy_id" column and perform aggregation operations on other columns.
- For numerical variables, we calculate statistics such as mean, median, mode, min, and max within each therapy_id group.
- For categorical variables, we apply another grouping based on therapy_id and calculate total unique counts for each class, as well as statistics like min, max, and mode.
- This process yields an enriched Rx_Claims data set with approximately 1,160 rows and an increased number of columns (in our case, we derived over 60 new variables).

2. Med_Claims Feature Engineering:

- In contrast, the Med_Claims data set presents 530 unique therapy_ids with a total of around 700 observations. Similarly, we apply feature engineering by grouping based on "therapy_id" and executing a set of aggregation operations.
- The result is a transformed Med_Claims data set with approximately 530 rows and an even larger number of new variables (in our case, over 110 variables).

Merging and Handling Null Values:

- After feature engineering for both Rx_Claims and Med_Claims, we have two enriched datasets.
- We merge these two data sets into one based on the therapy_id.
- Subsequently, we merge this combined data set with the Target data set using the target dataset's therapy_id.
- This process results in a comprehensive data set that incorporates all the derived variables from Rx_Claims, Med_Claims, and Target datasets.

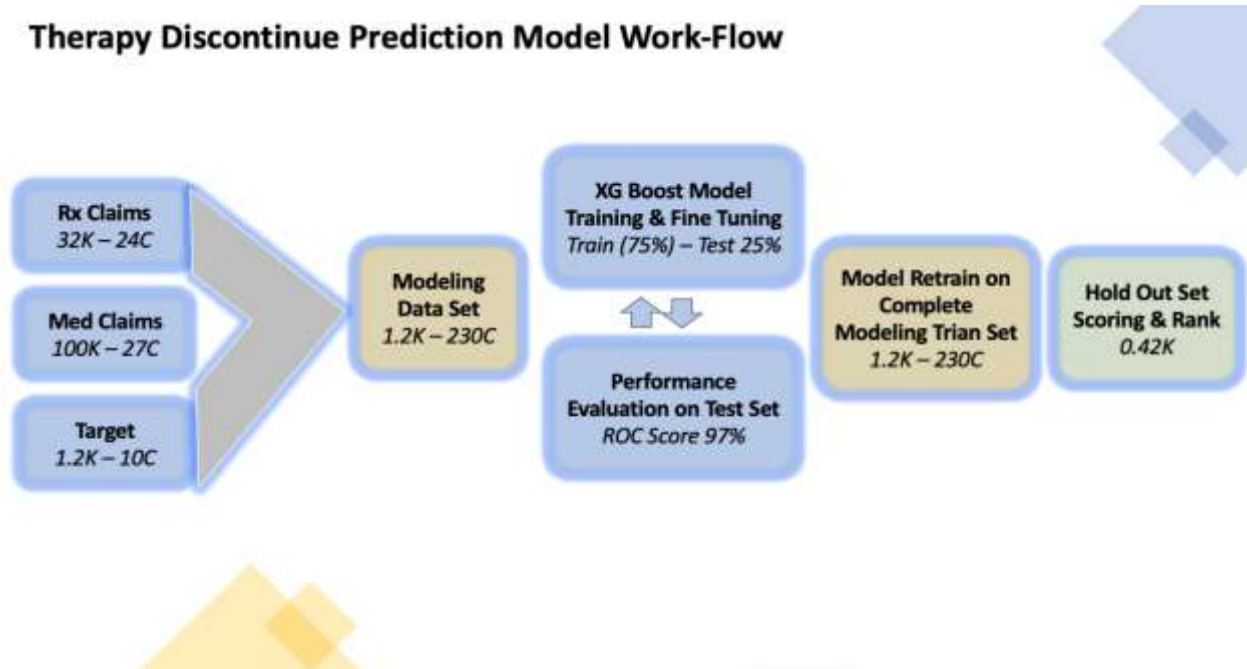
However, in this merging process, it's expected that some null values will emerge:

- The Rx_Claims dataset may contain around 70 null values for each column.
- The Med_Claims dataset may contain over 700 null values for each column.

To address these null values, we've chosen to impute them with "0." Essentially, this indicates that there was no relevant data for those specific therapy_ids or that those therapy_ids didn't have any corresponding claims.

The final outcome is a single data set that encompasses all the newly engineered variables from Rx_Claims, Med_Claims, and Target datasets. Detailed information about the workflow can be found in the model's workflow documentation.

5.3 Therapy Discontinue Prediction Model Workflow:



This comprehensive model workflow provides a clear and detailed view of the process we followed in developing our Machine Learning model. It encapsulates each critical step, from our approach to handling individual data sets to the final scoring and ranking of the Hold Out Set.

1. Data Set Preparation:

- We initiated the model development by preparing three distinct data sets: Target.csv, Rx_Clms, and Med_Clms. Each data set varied in terms of observations and columns.

2. Merging Data Sets:

- In order to construct a unified model, it was essential to merge these three data sets. We identified a common primary key, "therapy_id," that existed in all data sets. This key served as the linchpin for merging the data cohesively.

3. Feature Engineering:

- A pivotal phase of model development, feature engineering, was executed meticulously for each data set.
- *Rx_Claims Feature Engineering:*
 - In the Rx_Claims data set, which contained approximately 32,000 observations with 1,160 unique therapy_ids, we conducted aggregations on various columns.
 - For numerical variables, we computed key statistics such as mean, median, mode, min, and max within each "therapy_id" group.

- For categorical variables, we grouped them based on "therapy_id" and calculated statistics like total unique counts for each class, along with min, max, and mode.
 - The result was an enriched Rx_Claims data set with over 60 new variables.
- *Med_Claims Feature Engineering:*
 - In the Med_Claims data set, which included 530 unique therapy_ids with approximately 700 observations, similar feature engineering was applied. Aggregations and transformations were carried out to generate new variables.
 - This process yielded an enriched Med_Claims data set with more than 110 new variables.

4. Merging Enriched Data Sets:

- The enriched Rx_Claims and Med_Claims data sets were merged into one combined data set. This merging was carried out based on the common "therapy_id."

5. Merge with Target Data Set:

- Subsequently, this merged data set was further integrated with the Target data set. The therapy_id in the Target data set facilitated this merging process.

6. Handling Null Values in Modeling Data Set:

- In the merging process, it was expected that null values might arise. Specifically, the Rx_Claims dataset may contain around 70 null values for each column, while the Med_Claims dataset may have over 700 null values per column.
- To address these null values, a decision was made to impute them with "0." This signifies that the respective therapy_ids either lacked relevant data or had no associated claims.

7. Scoring and Ranking the Hold Out Set:

- The model workflow culminated with the scoring and ranking of the Hold Out Set, which involved applying the developed Machine Learning model to new data.
- The model's performance in scoring and ranking this data was a crucial evaluation of its effectiveness and predictive capabilities.

This model workflow serves as a comprehensive and structured guide to understanding the journey from initial data preparation to model development and validation. It illustrates our systematic approach to tackling the intricacies of the datasets and highlights the crucial role of feature engineering in creating an effective predictive model.

5.4 Model Evaluation and Performance Analysis:

Our model development journey started with a focus on identifying the most effective machine learning classification model. To achieve this, we explored six different types of classification models:

- **Logistic Regression**
- **XG Boost Classifier**
- **CART (Classification and Regression Trees)**
- **SVM Classifier (Support Vector Machine)**
- **Bayes Model**
- **Ensemble Stacking**

In the ensemble stacking approach, we combined four models: Logistic Regression, SVM Classifier, KNN (K-Nearest Neighbors), and XG Boost Classifier. Stacking is a powerful technique that leverages the strengths of multiple models for improved predictive performance.

Model Selection and Fine-Tuning:

Following an initial round of model training, we conducted an evaluation to determine which model performed the best. Among the six models, XG Boost Classifier, CART, and the Ensemble Stacking approach displayed promising results, with XG Boost Classifier leading the way. As a result, we decided to proceed with XG Boost for further refinement.

Hyperparameter Tuning:

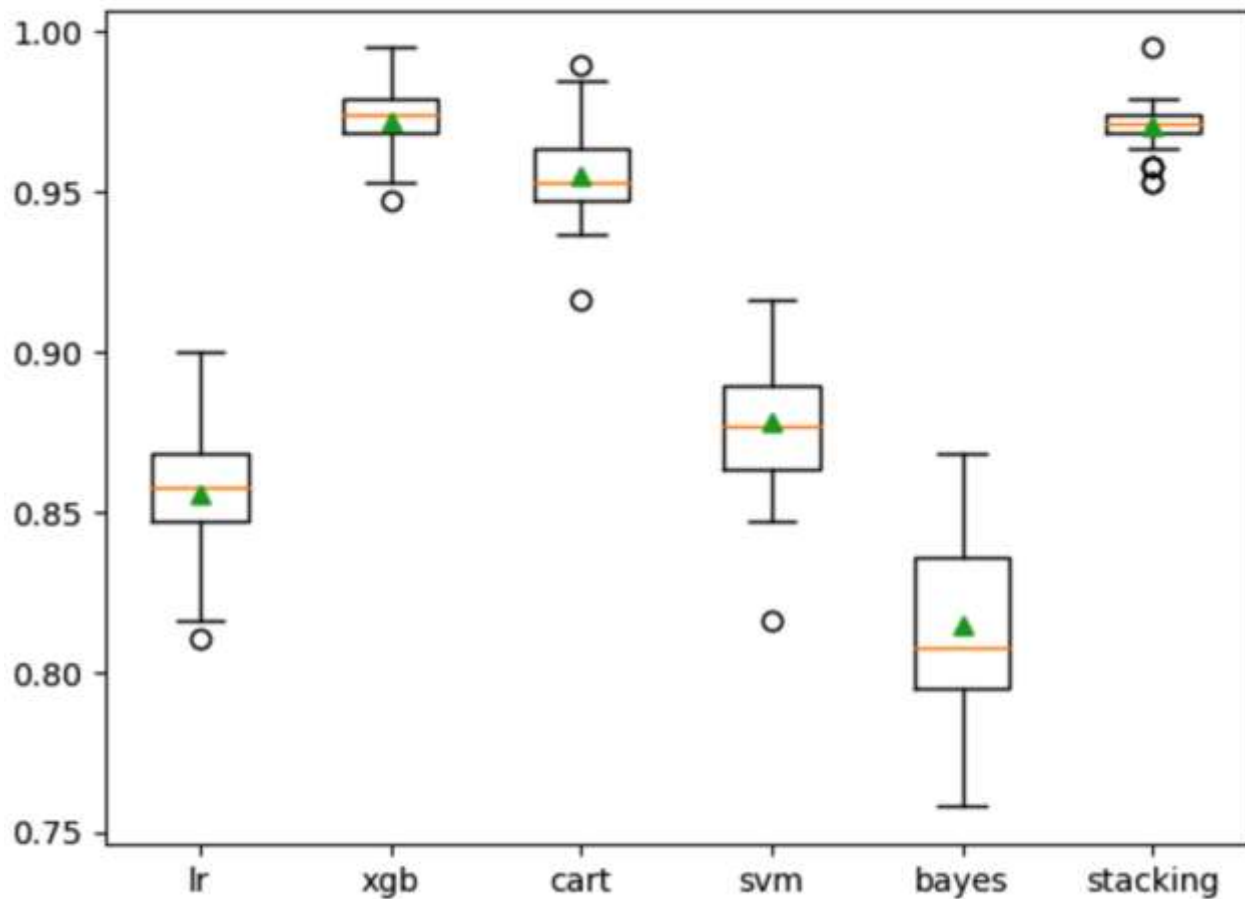
To optimize the XG Boost Classifier, we employed hyperparameter tuning using RandomizedSearchCV. This method helps in searching for the best combination of hyperparameters for the model. Once we conducted this process, we obtained the best hyperparameters that would enhance the model's performance.

Data Splitting and Evaluation:

For evaluating the XG Boost Classifier, we initially split the training data into two sets - a training set and a test set. This split was made with a 75:25 ratio, meaning 75% of the training data was allocated to the training set, while the remaining 25% constituted the test set. This approach is crucial for assessing the model's performance on unseen data.

Evaluation Metric:

The evaluation of the XG Boost Classifier model was based on the AUC (Area Under the Curve) score. AUC is a commonly used metric for binary classification models and measures the classifier's ability to distinguish between positive and negative classes. Achieving a high AUC score indicates a strong model performance.

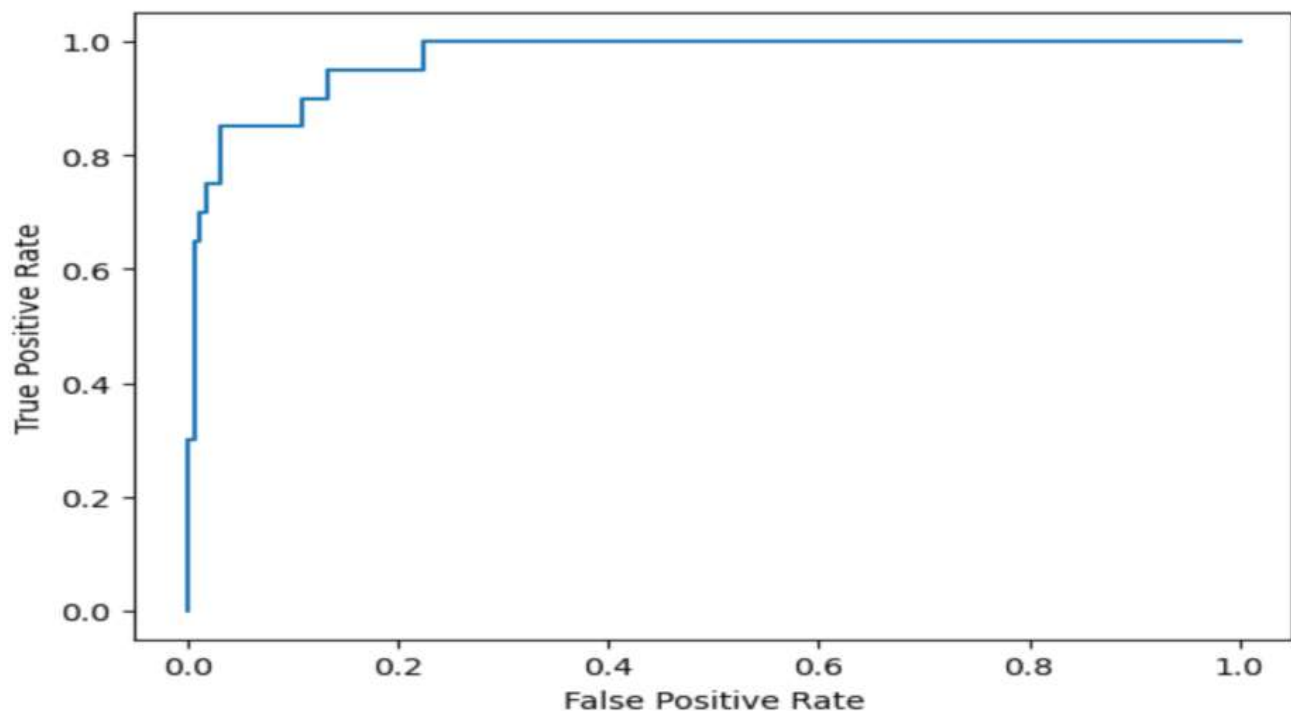


Performance Result:

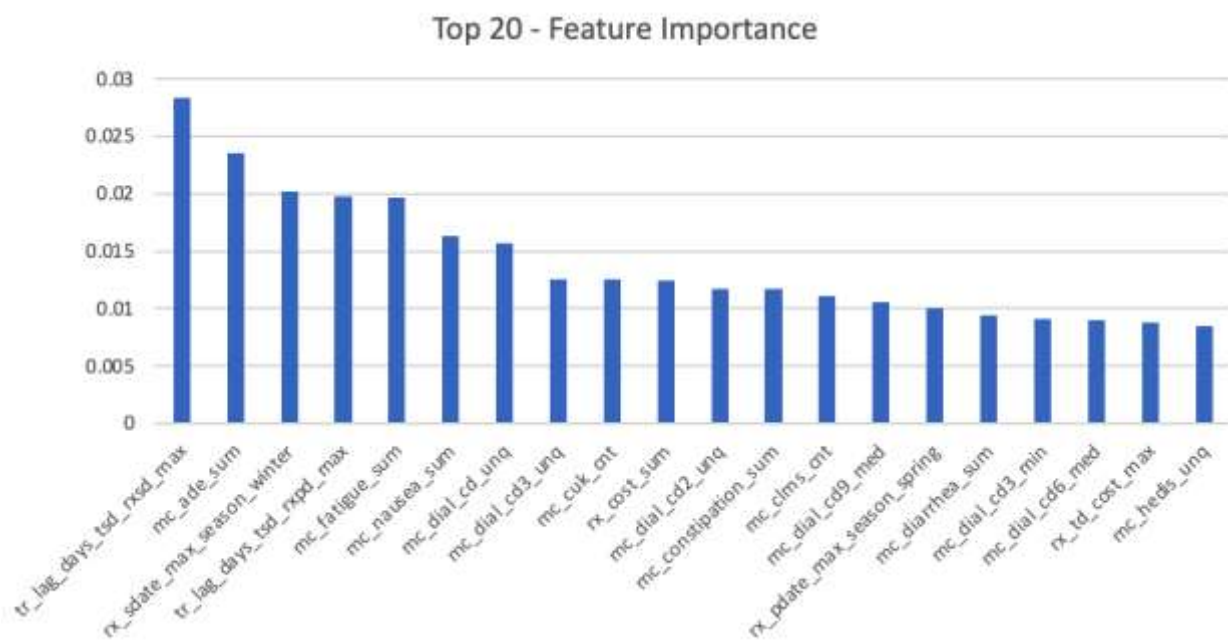
Upon evaluating the XG Boost Classifier, we obtained an impressive 97% AUC score on the test set. This signifies that the model excels in distinguishing between the target classes, demonstrating its robust predictive capabilities and making it a strong candidate for the final model choice.

In summary, our model development process began with the exploration of multiple classification models, and through systematic evaluation and fine-tuning, the XG Boost Classifier emerged as the model of choice. The successful hyperparameter tuning and the high AUC score on the test set underline the model's effectiveness in solving the classification task at hand.

ROC Curve of ADE Therapy Discontinue Classifier Model



5.5 Impactful Features for ADE Therapy Discontinuation Classifier Model



In our pursuit of understanding and addressing ADE (Adverse Drug Event) therapy discontinuation, it is imperative to identify the key features that contribute significantly to this phenomenon. Through extensive analysis, we have pinpointed the top 20 impactful features that play a crucial role in patients discontinuing their therapy during the specified period.

Top Feature: "tr_lag_days_tsd_rxsd_max"

Among these top contributing features, the variable "tr_lag_days_tsd_rxsd_max" stands out as the most influential. This variable's significance lies in its ability to shed light on the temporal aspect of ADE therapy discontinuation. "tr_lag_days_tsd_rxsd_max" is derived by calculating the lag days between the therapy start date from the target dataset and the pharmacy claim start date from the RX_Claims dataset.

Time Period as a Crucial Factor:

What sets "tr_lag_days_tsd_rxsd_max" apart is its revelation that not only the previously mentioned insights and recommendations impact ADE therapy discontinuation, but also the timing of when patients experience side effects and commence the claiming process plays a substantial role.

Elaborating on the Significance:

- **Temporal Awareness:** The "tr_lag_days_tsd_rxsd_max" variable is a powerful indicator of the temporal aspect of ADE therapy. It reveals the time gap between the therapy's initiation and when patients recognize their side effects and initiate the process of claiming these events.
- **Predictive Power:** This temporal awareness provides predictive power for identifying ADE therapy discontinuation. It enables healthcare professionals and policymakers to foresee potential discontinuation events and implement timely interventions.
- **Enhanced Patient Well-Being:** Understanding the time frame within which patients react to side effects and take action empowers us to tailor interventions that enhance patient well-being. It paves the way for timely support and improved therapy adherence.
- **Holistic Approach:** Combining the insights from various impactful features with temporal awareness creates a holistic approach to tackling ADE therapy discontinuation. It ensures that we account for not only patient characteristics but also the dynamic element of time in our efforts.

In conclusion, the "tr_lag_days_tsd_rxsd_max" variable, as the top impactful feature, highlights the significance of time in understanding and addressing ADE therapy discontinuation. It reinforces the notion that a comprehensive approach, considering both patient attributes and temporal dynamics, is essential in developing strategies and interventions to improve therapy adherence and overall patient outcomes.

6. Business Implication

6.1 Lung Cancer's burden on individuals, Family & Society

Lung cancer's impact on individuals, families, and society is a topic of significant concern as this impacts the socioeconomic fabric of an individual and a country. Below are some insights that can be interrelated to premature discontinuation of treatment.

Individuals:

1. **Health and Well-Being:** Lung cancer is known for its detrimental effects on an individual's health and overall well-being. Symptoms and consequences can be severe [1].
2. **Emotional and Psychological Impact:** A lung cancer diagnosis can cause considerable emotional distress, affecting mental health and well-being [5]. There by losing hope of survival
3. **Treatment Side Effects:** Lung cancer treatments often have debilitating side effects that can impact an individual's daily life [6].

Families:

1. **Emotional and Financial Strain:** Families of lung cancer patients often face immense emotional and financial burdens while providing care and support [11]. This can significantly reduce the family's productive earnings.
2. **Loss of Income:** Lung cancer patients may experience a loss of income due to their inability to work during treatment. This can strain family finances [1]. There by accumulating tendency to discontinue the treatment way before therapy end date.
3. **Coping with Grief:** Families have to cope with grief and loss if lung cancer is fatal, which can be a long-lasting and challenging process [12].

Society:

1. **Economic Impact:** Lung cancer imposes a substantial economic burden on society, including healthcare costs, productivity losses, and research and treatment development expenses [3].
2. **Public Health:** Addressing lung cancer's public health aspects involves anti-smoking campaigns, policy measures, and regulations to reduce tobacco use and environmental exposures [4].
3. **Research and Healthcare Infrastructure:** Society invests in lung cancer research, healthcare infrastructure, and support systems to better understand the disease and improve patient outcomes [7].

Lung cancer remains a complex public health issue, and addressing its multifaceted burden involves concerted efforts from individuals, families, and society as a whole, along with medical and public health organizations.

6.2 Lung Cancer treatment challenges

Lung cancer treatment poses several challenges to healthcare businesses like Humana. These challenges encompass financial, logistical, and patient care aspects. Here are some of the key challenges that healthcare businesses face when dealing with lung cancer treatment:

1. **High Treatment Costs:** Lung cancer treatment can be expensive, particularly for advanced stages of the disease. The cost of surgeries, chemotherapy, radiation therapy, targeted therapies, and immunotherapies can be a significant financial burden for healthcare businesses and insurers like Humana. Managing these costs while providing quality care is a challenge.
2. **Personalized Medicine:** Lung cancer is not a single disease; it comprises various subtypes with different genetic profiles. Personalized medicine, which tailors treatments to individual patients based on their genetic characteristics, has become a standard of care. Implementing and covering these advanced treatments while ensuring cost-effectiveness can be challenging.
3. **Access to Specialized Care:** Ensuring that lung cancer patients have access to specialized cancer centers and experienced healthcare providers is crucial. Healthcare businesses need to establish networks of specialists and ensure that patients can easily access the care they need.
4. **Early Detection and Screening:** Lung cancer is often diagnosed at advanced stages, making it harder to treat. Encouraging early detection through screening programs presents a challenge in terms of patient engagement and cost-effectiveness.
5. **Managing Side Effects:** Lung cancer treatments can have significant side effects that impact a patient's quality of life. Healthcare businesses need to address the management of side effects and ensure patients receive appropriate supportive care.
6. **Clinical Trials:** Participation in clinical trials is vital for advancing lung cancer treatment. Healthcare businesses like Humana may need to support and cover the costs of clinical trial participation for eligible patients.
7. **Patient-Centered Care:** Providing patient-centered care that addresses the physical, emotional, and psychological needs of lung cancer patients is a challenge. Ensuring that

patients and their families have the information and support they need is essential.

8. **Data Management:** Managing vast amounts of patient data, including medical records and genetic information, is a challenge. Ensuring data privacy and security while using this information for treatment decisions is a key concern.
9. **Preventive Initiatives:** Encouraging preventive measures such as smoking cessation and reducing exposure to environmental carcinogens is part of lung cancer control. Healthcare businesses can play a role in these public health initiatives.
10. **Policy and Regulatory Changes:** Changes in healthcare policies and regulations can impact the reimbursement and coverage of lung cancer treatments. Healthcare businesses must adapt to evolving policies.
11. **Cultural and Socioeconomic Factors:** Lung cancer disproportionately affects certain populations. Understanding and addressing cultural and socioeconomic factors that impact access to care is essential for equitable treatment.

Lung cancer treatment challenges are complex and multifaceted, and healthcare businesses like Humana need to navigate these challenges while ensuring that patients receive optimal care and support. Collaborative efforts among healthcare providers, insurers, policymakers, and patient advocacy groups are crucial to address these challenges effectively.

6.3 Hypothesis Analysis on current believed cause of Discontinuation.

As provided in the case study the primary reason for the therapy discontinuation is the side effects of the drug Osimertinib, but there're are various other factors that are responsible for the discontinuation of the therapy.

There are several reasons why some NSCLC patients may discontinue or drop out from therapy:

1. **Disease Progression:** In some cases, despite treatment, the cancer continues to progress, leading to a lack of response to therapy. Patients may become discouraged and opt to discontinue treatment when the disease becomes advanced and incurable.
2. **Financial Burden:** The cost of cancer treatment, even with insurance coverage, can be significant. Some patients may discontinue treatment due to financial stress, such as high co-pays, deductibles, and the inability to work during therapy.
3. **Quality of Life:** Patients may prioritize their quality of life over aggressive treatments. They may choose to discontinue therapy to have more time without the side effects and constraints of treatment.

4. **Logistical Challenges:** Some patients may face logistical challenges, such as difficulties in accessing treatment centers or transportation issues, which can lead to discontinuation.
5. **Lack of Social Support:** Patients without strong social and emotional support systems may be more likely to discontinue therapy. Support from family and friends can play a crucial role in helping patients continue with treatment.
6. **Psychological Impact:** A lung cancer diagnosis can have a significant psychological impact. Some patients may experience depression, anxiety, or distress, which can lead to treatment discontinuation.
7. **Physician-Patient Communication:** Effective communication between patients and their healthcare providers is critical. If patients do not fully understand their treatment plans or do not feel they are receiving adequate support, they may discontinue therapy.
8. **Comorbidity:** Some NSCLC patients may have other health issues or comorbidities that complicate their ability to tolerate treatment.
9. **Clinical Trial Participation:** Some patients may discontinue standard treatment in favor of enrolling in clinical trials to explore experimental treatments and potentially better outcomes.

It's important to note that discontinuing treatment is a complex and deeply personal decision for cancer patients. While some patients may choose to discontinue therapy for the reasons mentioned above, others may continue treatment and explore alternative therapies. Effective patient-physician communication, support systems, and personalized treatment plans can help patients make informed decisions about their care. It's crucial for patients to discuss any concerns or side effects with their healthcare providers and to explore all available options to optimize their treatment and quality of life.

6.4 Recommendations for ADE Therapy Discontinuation:

Based on Exploratory Data Analysis:

- **Gender Disparities:** Investigate gender-based disparities in therapy discontinuation. Develop gender-specific interventions to enhance therapy adherence and satisfaction.
- **Racial Disparities:** Examine racial distribution among patients who discontinue therapy. Implement strategies to reduce discontinuation rates, particularly among racial groups with higher rates.
- **Treatment Location Coordination:** Improve the coordination and experience for patients selecting multiple treatment locations during ADE therapy. Ensure consistency and high-quality care across different locations.
- **Complex Interactions:** Explore the interplay of factors such as race, gender, and side effects in therapy discontinuation. Consider comprehensive analysis to address disparities among different patient groups.
- **Financial Barriers:** Recognize the impact of income on therapy discontinuation. Tailor interventions and support systems to mitigate financial barriers and improve therapy adherence.
- **Disability and Gender Dynamics:** Investigate relationships between disability, gender, and therapy outcomes. Develop tailored interventions to enhance therapy adherence and patient outcomes.

Based on Hypothesis Analysis:

- **Disease Progression:** Implement strategies for closely monitoring advanced and incurable cases. Offer alternative therapies or palliative care for those with progressing diseases.
- **Financial Support:** Create mechanisms to alleviate the financial burden on patients. Explore options for reducing costs, offering financial assistance, or connecting patients with relevant resources.
- **Quality of Life:** Engage in open discussions with patients about their treatment preferences. Offer symptom management and palliative care to improve quality of life.
- **Logistical Solutions:** Address logistical challenges by providing transportation assistance or telehealth options for patients who struggle to access treatment centers.
- **Social and Emotional Support:** Establish support programs for patients lacking strong social systems. These programs can provide emotional and practical assistance to improve therapy adherence.
- **Psychological Counseling:** Offer psychological support services to help patients cope with the emotional impact of a lung cancer diagnosis. Identify and treat depression, anxiety, or distress promptly.
- **Communication Improvement:** Focus on improving communication between healthcare providers and patients. Ensure that patients fully understand their treatment plans and have access to the support they need.
- **Comorbidity Management:** Develop tailored treatment plans for patients with comorbidities. Consider alternative therapies or supportive care when necessary.

- **Clinical Trial Information:** Provide comprehensive information about available clinical trials, ensuring that patients make informed decisions regarding their treatment.

Based on Modeling Results:

- **Temporal Awareness:** Recognize the importance of time in therapy discontinuation. Use the "tr_lag_days_tsd_rxsd_max" variable to predict and prevent discontinuation events proactively.
- **Predictive Power:** Leverage temporal awareness to predict discontinuation risks and intervene promptly. Develop data-driven strategies to transform patient care.
- **Enhanced Patient Well-Being:** Use temporal insights to provide timely support, enhancing patient well-being and therapy adherence.
- **Holistic Approach:** Combine patient attributes and temporal dynamics for a comprehensive approach to tackling discontinuation. Focus on both patient characteristics and the dynamic element of time.

These recommendations, based on hypothesis analysis, exploratory data analysis, and modeling results, offer a multifaceted approach to addressing ADE therapy discontinuation. They emphasize early intervention, patient-centered care, and support for vulnerable populations, ultimately improving patient outcomes and adherence.

7. Conclusion

The primary challenge we face is enhancing patient adherence to Osimertinib, a life-saving medication, while effectively managing its side effects. Non-adherence in the initial six months, particularly among those experiencing side effects, poses a substantial hurdle, jeopardizing patient safety and squandering the medication's life-saving potential.

To tackle this issue, we must harness data and analytics to pinpoint Humana members at risk of discontinuing Osimertinib due to side effects. By implementing adherence strategies, offering counseling, and providing support to manage side effects, our objective is to empower oncology patients for healthier and longer lives. The key lies in creating a data-driven intervention plan, identifying at-risk individuals, guiding them, and ultimately enhancing adherence rates and health outcomes.

Tolerance of Osimertinib can be challenging due to side effects like nausea, fatigue, and pain. Many patients may discontinue treatment rather than seeking guidance. About one quarter of Humana members discontinue Osimertinib therapy due to side effects in the first 6 months.

In our modeling process, we identified the XG Boost Classifier as the most effective model, achieving a 97% AUC score after hyperparameter tuning. Extensive feature engineering on RX-Claims and Med-Claims datasets provided a robust modeling dataset, improving patient outcomes.

Our recommendations include addressing gender and racial disparities, improving treatment coordination, understanding complex interactions, mitigating financial barriers, and enhancing disability and gender dynamics based on exploratory data analysis. Hypothesis analysis suggests strategies for disease progression monitoring, reducing financial stress, prioritizing quality of life, addressing logistical challenges, and offering social and emotional support. Modeling results emphasize the temporal aspect's significance and advocate for a holistic approach that combines patient attributes and temporal dynamics. These recommendations collectively offer a multifaceted approach to enhance therapy adherence, ultimately improving patient outcomes.

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